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Systematic review

Resistance training, skeletal muscle hypertrophy, and glucose homeostasis: how related are they? A Systematic review and Meta-analysis

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Abstract

Resistance training (RT) promotes skeletal muscle (Skm) hypertrophy, increases muscular strength, and improves metabolic health. Whether changes in fat-free mass (FFM; a surrogate marker of muscle hypertrophy) moderate RT-induced improvements in glucose homeostasis has not been determined, despite extensive research on the benefits of RT for health and performance. The aim of this meta-analysis is to examine whether RT-induced Skm hypertrophy drives improvements in glucose metabolism and to explore confounders, such as biological sex and training parameters. Random-effects meta-analyses were performed using variance random effects. Meta-regressions were performed for confounding factors depending on the heterogeneity (I^2). Analyses from 33 intervention studies revealed significant within-study increases in FFM with a moderate effect size (within-studies: (effect size; ES = 0.24 [0.10; 0.39]; $p = 0.002$; $I^2 = 56\%$) and a tendency for significance when compared with control groups (ES = 0.42 [−0.04–0.88]; $p = 0.07$). Within-study significant increases in glucose tolerance (2 h glucose: ES = −0.3 [−0.50; −0.11]; $p < 0.01$; $I^2 = 43\%$; glucose area under the curve (AUC): −0.40 [−0.66; −0.13] $I^2 = 76.1\%$; $p < 0.01$) and insulin sensitivity (ES = 0.38 [0.13; 0.62]; $I^2 = 53.0\%$; $p < 0.01$) were also apparent with RT. When compared to control groups, there was no significant difference in 2 h glucose, nor in glucose AUC from baseline in RT intervention groups. Meta-regression analyses failed to consistently reveal increases in FFM as a moderator of glucose homeostasis. Other mixed-effect models were also unsuccessful to unveil biological sex or training parameters as moderators of FFM increases and glucose homeostasis changes. Although Skm hypertrophy and improvements in glycemic control occur concurrently during RT, changes in these variables were not always related. Well-controlled trials including detailed description of training parameters are needed to inform RT guidelines for improving metabolic health. Registration and protocol number (Prospero): CRD42023397362.

Key words: insulin sensitivity, glucose tolerance skeletal muscle mass, muscle quality, weight training, muscle mass gain, fat-free mass gain

1. Introduction

The benefits of resistance training (RT) for athletic performance and well-being across health and disease are well established in the exercise physiology literature (Kraemer et al. 2002; Westcott 2012). Extensive research has suggested that RT, by enhancing skeletal muscle (Skm) metabolism, is a potent strategy to prevent or even treat metabolic diseases, such as the metabolic syndrome, insulin resistance (IR), and type 2 diabetes (T2D) (Colberg et al. 2010). Some of these benefits are distinct from those induced by aerobic exercise, such as increasing strength, and promoting Skm hypertrophy (Schoenfeld et al. 2017*a*; Grgic et al. 2018). Indeed, recognized exercise physiology and sports medicine organizations recommend that adults ≥ 18 years old perform RT at least 2 days per week, using loads varying from 30%–50% to 80% of maximal strength (Ratamess et al. 2019; Fragala et al. 2019; Ross et al. 2020).

The effects of RT interventions on glucose homeostasis have been extensively studied in the past decades. Furthermore, RT has been shown to improve clinical glucose homeostasis parameters, measured under experimental metabolic challenges, such as glucose tolerance assessment with an oral glucose tolerance test (OGTT) (Molsted et al. 2013; Frank et al. 2016) and direct measurement of insulin sensitivity with the euglycemic-hyperinsulinaemic clamp (CLAMP) (Andersen et al. 2003; Holten et al. 2004; Shaibi et al. 2006). Because Skm is thought to account for up to 80% of insulin-dependent glucose uptake (Defronzo and Tripathy 2009), numerous studies have promoted the idea that quantitative changes in fat-free mass (FFM) following RT drive improvements in whole-body glucose homeostasis (Miller et al. 1984; Ferrara et al. 2006; Kleinert et al. 2013; Shepherd et al. 2014; Bucci et al. 2016; Kogure et al. 2016). One of the proposed mechanisms lies upon the assumption of a greater capacity for glucose storage

in the form of glycogen, since Skm tissue represents most of the glycogen pool in the body (Knuiman et al. 2015). On the other hand, prospective cohort studies have related the development of metabolic impairments with a decrease in Skm mass, notably with aging (Srikanthan and Karlamangla 2011; Lee et al. 2018). Therefore, researchers in the field of exercise physiology and especially in the RT area have shown an increased interest in Skm hypertrophy as a potential health enhancer (Westcott 2012; Joanisse et al. 2020).

However, accumulating evidence challenges this paradigm (Kuk et al. 2008; Barsalani et al. 2013; Ghachem et al. 2019; Mesinovic et al. 2019). Furthermore, evidence providing mechanistic explanation, such as linking RT-induced Skm hypertrophy and enhanced mitochondrial function with peripheral glucose uptake, is currently scarce (Groennebaek and Vissing 2017; Paquin et al. 2021). Furthermore, although this paradigm is well accepted and applied in the exercise physiology field, there appears to be no statistical evidence to support it, to the best of our knowledge. Hence, the main purpose of this meta-analysis is to assess the magnitude of both Skm hypertrophy/FFM increase and glucose homeostasis improvement in response to RT interventions in the general/adult population. A second objective was to assess the relationship between RT-induced FFM gain and improvements in glucose homeostasis.

2. Methods

2.1. Experimental approach to the problem

This meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al. 2021).

2.1.1. Study design, population, and inclusion and exclusion criteria

Population, intervention, comparator/control, outcomes, and study design approach were used to determine the boundaries within which the review was performed. Participants included the general adult population (≥18 years old) with no limit of age. Studies were excluded if they were conducted in rodents, pediatric populations, and cardiac rehabilitation programs; they included participants with type 1 diabetes, HIV-treated patients, or were undergoing hormonal therapy because of the known effect of medication on glucose homeostasis and/or lean tissue.

2.1.2. Intervention and comparator/control

Research articles were considered eligible for this meta-analysis if they included an RT intervention of a duration of at least four (4) weeks and included a minimum of two (2) sessions per week (Lim et al. 2022). Studies involving mixed-exercise interventions (aerobic combined with RT, or circuit-type RT that significantly solicits the aerobic system, involving ≥10 min aerobic warm-up or cool down), or included another intervention that may affect glucose homeostasis (e.g., nutritional intervention) were excluded. Only studies including whole-body RT were considered eligible. No exclusion criteria were established based on intensity or volume, to have the most accurate portrait of current literature. Aerobic training, because of its recognized impact on glucose homeostasis, was not considered as a control or comparator.

2.1.3. Outcomes

Primary outcomes of interest were changes in glucose homeostasis and Skm/FFM. Direct measurement of Skm hypertrophy in response to RT comprises several methodological challenges, such as cost, invasiveness, and technical resources and knowledge, especially for methods such as magnetic resonance imaging (MRI) and computed tomography. Consequently, FFM estimation methods have been developed to limit aforementioned challenges. These tools, especially hydrodensitometry, air-displacement plethysomography, MRI, and dual X-ray absorptiometry (DXA-scan), are considered as high validity and reliability methods (Fosbøl and Zerahn 2015; Kuriyan 2018). Hence, FFM changes are widely used in exercise studies as a surrogate of Skm hypertrophy with a great level of confidence. Only studies including either DXA-scan, hydrodensitometry, MRI, or air displacement plethysmography were included. Studies using bioimpedance and skinfold calipers were excluded, due to their lessened validity for actual body composition determination (Kuriyan 2018). For glucose homeostasis determination, only studies including a metabolic challenge, i.e., OGTT, CLAMP, or intravenous glucose tolerance test (IVGTT) were included. The rationale for this choice was based on the assumption that these methods provide direct or estimated peripheral glucose uptake (Abdul-Ghani et al. 2007).

2.1.4. Study designs included

Randomized controlled trials (RCTs), non-RCT intervention studies, randomized parallel-arm, single-arm, and pilot studies were considered for analysis. Inclusion of studies with designs other than RCT was considered eligible mostly because conducting RCTs in the state-of-the-art approach is not always feasible in exercise science settings (Hecksteden et al. 2018). Studies were included if they also were published in a peered review, published in English or French, had primary or secondary objectives of assessing changes in body composition and glucose homeostasis in response to RT, and were using RT in isolation (without any other intervention) in at least one experimental group.

2.2. Search strategy

Publication research was last performed on 23 September 2023 in EBSCO using five different databases: Academic Search Complete, SPORTDiscuss, CINAHL, MEDLINE, and AMED. Restrictions were applied for language (English and French only) and for publication period (from January 1960 to September 2023). The search protocol for the current meta-analysis was registered in the University of York International prospective register of systematic reviews (PROSPERO; ID: CRD42023397362). Ryaan (https://www.rayyan.ai/) (Mourad

Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines flow chart for literature search and study selection.

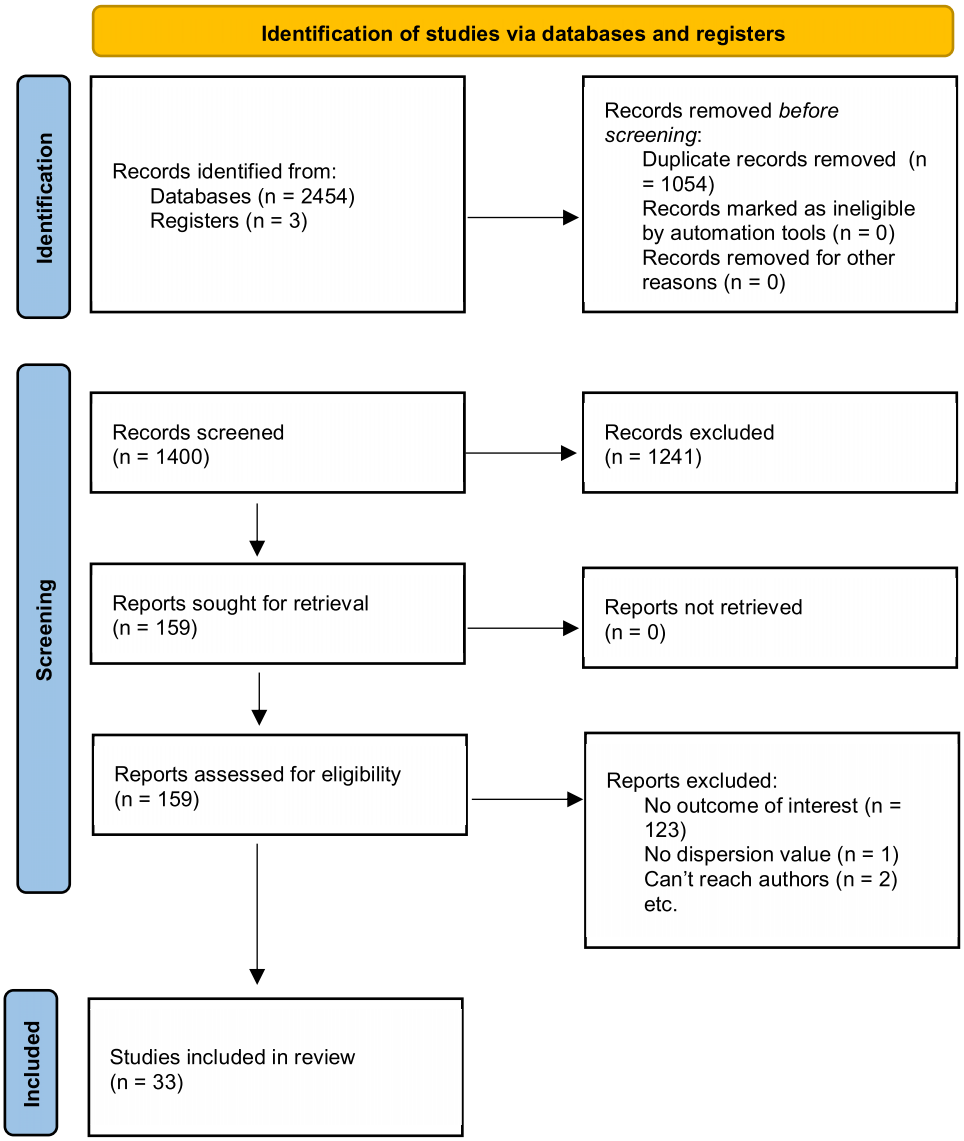

Table 1. Keywords used for the search strategy.

Intervention	"resistance training" OR "resistance exercise" OR "strength training" OR "strength exercise" OR "strength program" OR "weight training" OR "weight exercise" OR "weight lifting" OR "resistive training" OR "resistive exercise" OR "strengthening program" OR "strengthening exercise"
	AND
Outcomes	"insulin sensitivity" OR "insulin resistance" OR "oral glucose*" OR "glucose tolerance" OR "glucose homeostasis" OR "glucose metabolism" OR "mixed meal tolerance test" OR "ogtt" OR "mmtt" OR "mixed meal challenge test" OR "glucose disposal*"

Note: The research was performed in EBSCO Host on 23 September 2023 including the following database: Academic Search Complete, AMED, CINA HL, Medline, and Sportdiscuss.

Ouzzani 2016) was manually used (no automatic screening) to record decisions based on Title/Abstract screening. Finally, we performed a cross-referencing literature search, i.e., we consulted all meta-analyses and reviews on related field/research question (Greenhalgh and Peacock 2005) to yield papers that may have been missed by database search. Two additional studies were identified as eligible through this strategy. For further details, PRISMA flow diagram is presented in Fig. 1. The final search equation is displayed in Table 1.

2.3. Selection and data collection process

Final study selection was performed by two independent teams comprising two reviewers for a total of four investigators (JP/HI and AMC/RT). Each investigator individually

reviewed the full text of the selected studies for further inclusion or exclusion. Reasons for inclusion or exclusion were recorded in an Excel spreadsheet and were compared between the two investigators before a final decision was taken. Disagreements between reviewers were resolved by two other independent co-authors (IJD/ER). After study selection, the two teams independently collected data using the same strategy as in study selection. In the case of essential data missing or lacking in selected studies, authors were e-mailed by one of the two independent co-authors.

2.4. Data extraction

The following data were extracted: study country, study design, total number of participants (*N*), total number of participants' data analyzed, number (*n*) of male participants, number (*n*) of female participants, % female participants, method of glucose homeostasis assessment, and method of body composition assessment were collected as study characteristics. Whenever available, participant's characteristics were extracted as age (years), height (m), weight (kg), BMI (kg/m^2), smoking status, fasting glucose metabolism (A1c % and/or fasting insulin and/or fasting glucose), number of participants using medication and type, presence of T2D, T2D duration, general RT-exercise modalities (free weights, machines, TheraBand, etc.), RT-exercise prescription parameters (number and names of exercises, number of sets, repetitions, eccentric/concentric, frequency/week, intensity as % one-repetition-maximum (1RM), number of sets per session, rest time between sets, and session duration in min), intervention duration, number of sessions, and load adjustment during intervention. Main outcomes were extracted as follows: body composition data were collected as absolute FFM and FM values (kg) pre- and post-intervention; results from glucose homeostasis/insulin sensitivity were collected as 2 h glucose (mmol/L) and/or glucose AUC (mmol/L*120 min or mg/dL*120 min*10^{-3}) and/or glucose infusion rate (mg/kg/min or mg/kgFFM/min) and/or glucose rate of disappearance (Rd/μmol/kg/FFM/min) pre- and post-intervention. The Plot digitizer online app (https://plotdigitizer.com/app) was used for the extraction of raw data from graphical results.

3. Statistical analyses

3.1. Effect measures/calculation of effect size

The main outcomes were changes in glucose homeostasis, defined as glucose tolerance and insulin sensitivity, and delta FFM. For each study, measures of changes between baseline and endpoint were collected as mean of the change and standard deviation of the change for each arm. Data for each variable were extracted as mean ± standard deviation (SD). When data were available as median or interquartile range (IQR), authors were contacted and asked for mean ± SD after which appropriate transformations were performed according to Cochrane guidelines. If there was no response from the authors, Wan equation (Wan et al. 2014) was used to transform median and IQR to mean and SD. When considering studies with a control arm, we presented pooled estimates for those continuous outcomes as standardized mean differences (SMDs) with 95% confidence interval (CI). Otherwise, when excluding control arm, we presented pooled estimates of mean of change using the generic inverse variance method. We used the R v.4.0.2 software to perform meta-analyses using random-effects models.

3.2. Synthesis methods

The effect of RT on glucose tolerance and insulin sensitivity was analyzed in R (*meta* package) using random-effects meta-analyses performed according to Cochrane guidelines. As it may be deemed unethical, or very difficult to conduct, a great deal of studies within the field of exercise physiology leaves out a control group. To provide a more thorough and realistic depiction of the current literature, we conducted analyses within individual studies (pre- vs. post-intervention) for all studies and included comparisons with a control group (no exercise) in articles that employed an RCT design. All included studies were first analyzed for SMD in FFM. Then, they were analyzed on separate mixed-effects models according to glucose homeostasis measurement. Results are presented in caterpillar forest plot based on FFM increase, meaning that the studies are presented in descending order from the greatest increase in FFM to the smallest. Heterogeneity between studies was assessed with the I^2 statistic and Cochran's Q test. When applicable, studies with a control group were analyzed separately. Low heterogeneity was defined as a I^2 of 25% or below, moderate heterogeneity between 25% and 50%, and high heterogeneity above 50%. We also assessed statistical heterogeneity by χ^2 tests ($p < 0.10$). Meta-regressions were performed for confounding variables (e.g., magnitude of FFM change, proportion of females in study sample, training intervention duration, fat mass (FM) changes (delta kg), changes in VO_2max/VO_2peak, total volume (defined as total number of sets performed across the intervention), and average age of sample) depending on heterogeneity (I^2). Subgroup differences were analyzed in mixed-effects models with studies comprising T2D participants as a sensitivity analysis. Finally, a tendency toward significance was defined as $p \leq 0.100$ (Thiese et al. 2016; Ganesh and Cave 2018).

3.3. Assesment of risk of bias

The assessment of risk of bias has been done at the study level by two independent teams of two authors (JP/HI and AMC/RT) using the PEDro (Physiotherapy Evidence Database (Macedo et al. 2010)) scale for a maximal score of 11 points. Each of the following criteria was screened: specification of eligibility criteria, randomization, concealed allocation, groups similar at baseline, blinding of all subjects, blinding of all exercise specialists/medical staff, at least one key outcome was obtained in a minimum of 85% of participants allocated to group, between-group statistical comparisons, or intention to treat. No studies have been excluded based on their PEDro scale score. The quality of the included studies was then defined according to the following categories: "excellent" 9–11 points; "good" 6–8 points; "fair" 4–5 points; and "poor" ≤3 points (Macedo et al. 2010). Any study with a PEDro scale ≤ 3 was considered as having a high risk of bias. Any

disagreements were resolved through discussion between the authors until a consensus was reached or by two independent coauthors (ER/IJD).

3.4. Assessment of certainty of evidence using the GRADE approach

Each review outcome (outlined as each statistical model undertaken) certainty of evidence was assigned using a quality rating based on the Grading of Recommendations Assessment Development and Evaluation (GRADE) working group model (Balshem et al. 2011). This strategy allows an estimation of the level of confidence in the certainty of evidence offered by statistical models undertaken, according to five domains, i.e., risk of bias of individual studies, heterogeneity, imprecision, likelihood of publication bias, and directedness. Each individual study was a priori evaluated based on type of study design (RCT or other), PEDro scale-quality level (risk of bias), directedness of the evidence provided, and presence of dose-response or substantial effect (https://training.cochrane.org/handbook). Each outcome was then examined based on individual studies included. Outcome comprising evidence mostly from non-RCTs was a priori graded as low certainty of evidence, and then upgraded or downgraded based on imprecision (sample size and CI), overall risk of bias, and inconsistency (heterogeneity). Outcome comprising evidence mostly from RCTs was *a priory* graded as high certainty of evidence, and then upgraded or downgraded based on overall risk of bias, inconsistency, and presence of dose-response or substantial effect (Balshem et al. 2011).

4. Results

4.1. Study selection and characteristics

Table 2 displays the studies identified and selected by the search strategy. A total of 2454 studies were initially identified through database searching. A total of 1400 studies remained after duplicates were removed, and 1241 were excluded based on Title/Abstract screening. A total of 159 were assessed for eligibility by full text screening, from which 126 were excluded (see Fig. 1 for further details). Hence, a total of 33 studies included 1054 participants for analyses, among whom 48% were female and 52% were male. The date of publication ranged from 1984 to 2023 with more than half of studies (19/33) conducted in the USA. The duration of RT interventions ranged from 6 to 36 weeks with an average frequency of 3.1 ± 0.7 times per week, using a somewhat wide range of methodologies, with 3 × 10–12 repetitions at 80% of one-repetition-maximum (1RM) being the most frequent (approximately 50% of studies, depending on how details are provided). Full details on anthropometrics, age, and exercise interventions such as Frequency, Type, Time, Volume, and Progression prescription (Bushman 2018) can be found in supplemental material (Table S1). Selected studies displayed high quality methods for glucose tolerance (OGTT) or peripheral insulin sensitivity tests (CLAMP) and for body composition assessment (with one out of three studies having used DXA-scan). Importantly, only one study (Poehlman et al. 2000) using a CLAMP included a control group. Therefore, only within-studies analyses were performed for CLAMP studies. Moreover, only one study using IVGTT was considered eligible with regard to other inclusion/exclusion criteria. The assessment of risk of bias and certainty of evidence at the level of the individual study (GRADE) is presented in Table S1 (supplementary files) and for each outcome assessed in Table 3. Of 33 selected studies, 36% were graded as being of good or excellent quality, with 9% (three studies) achieving high quality (≥9). Most studies (57%) were graded as having fair quality (between 4 and 5) and three studies were graded as having poor quality (≥3). Overall certainty of evidence ranges from very low to moderate based on GRADE criteria. Figure 2 displays funnel plots for assessment of risk of publication bias for 2-h glucose and CLAMP studies.

4.2. Results of analyses

Analysis on pre–post intervention shows SMD of individual studies' pre–post-intervention SD of change only (within-group SDs), whereas analysis on RT and control groups comparisons show intervention versus control group differences in SD of change from baseline. Results are presented in forest plots in descending order of FFM increases (from most to least) and are shown as SMD [95 CI%]; p-value; and I^2 for heterogeneity. Table 3 summarizes SD of change (noted as relative effect) as well certainty of evidence (GRADE) evaluated as previously described. Figures illustrating results from within-studies analyses can be found in supplementary material.

4.3. Fat-free mass quantitative changes following RT interventions

4.3.1. Analysis on pre–post intervention

Within-studies analyses (pre vs. post intervention, including single-arm and parallel-arm studies) revealed a significant increase in FFM with a moderate effect size (0.24 [0.10; 0.39]; $p < 0.01$; $I^2 = 56\%$), which represents a mean FFM increase of 1.2 ± 0.8 kg.

4.3.2. Analyses on studies including a control group

When restricting analyses to studies including a control group, a tendency for increasing FFM was observed (0.42 [−0.04; 0.88], $p = 0.07$; $I^2 = 81\%$; Fig. 3).

4.4. Changes in glucose homeostasis

4.4.1. Analysis on pre–post intervention

Within-studies SMDs revealed that RT improved 2 h glucose and glucose AUC in 15 out of 19 and 13 out of 16 studies, respectively. More specifically, we observed a significant improvement in 2 h glucose (−0.30 [−0.50; −0.11]; $p < 0.01$; $I^2 = 43\%$) and glucose AUC (−0.40 [−0.66; −0.14]; $I^2 = 76.1\%$; $p < 0.01$), both with moderate effects sizes. Within-studies analyses also revealed a significant improvement in insulin

Table 2. Overview of studies included in the meta-analysis.

Study	Main characteristics of participants	Age (years)	RT overview	Duration (weeks)	Glucose metabolism	BC measurement	Quality rating/11 (PEDro scale)
Baldi and Snowling (2003)	Individuals with T2D	Control: 50.1 ± 3.9 RT: 46.5 ± 6.3	3×/week, 12 reps per exercise for 10 exercises (intensity not specified)	10	OGTT	Hydrostatic weighting	5 (Fair)
Bobeuf et al. (2010)	Older, healthy individuals	Control: 66.3 ± 3.7 RT: 65.9 ± 3.2	F: 3×/week, 3 × 80% of 1RM for all exercises except 3 × 20 reps for abdominal exercises	24	IVGTT	DXA	6 (Good)
Bucci et al. (2016)	Offspring of lean/normal weight (OLM) or overweight/obese mothers (OLM)	OLM: 72.3 ± 2.7 OOM: 71.5 ± 3.7	3×/week, 8–15 reps at 50%–80% 1RM for eight exercises	16	CLAMP	MRI	5 (Fair)
Consitt et al. (2013)	Young and aged healthy individuals	Young: 23.6 ± 5.0 Aged: 69.3 ± 9.0	3×/week, 8–15 reps at 50%–80% 1RM	16	CLAMP	DXA	6 (Good)
Dionne et al. (2004)	Young and older healthy individuals	Young: 27.8 ± 3.5 Aged: 66.6 ± 4.9	3×/week, three sets at 80% 1RM	24	CLAMP	DXA	4 (Fair)
DiPietro et al. (2008)	Healthy, inactive older female	77 ± 6	4×/week with thera bands	36	OGTT	DXA	8 (Excellent)
Eikenberg et al. (2016)	Older individuals with impaired glucose metabolism	59.6 ± 5.4	2×/week, one set of 6–12 reps per exercise, “Hard” on the 6–20 borg scale.	12	OGTT	DXA	5 (Fair)
Ferrara et al. (2006)	Older male with normal and impaired glucose metabolism	50–79	3×/week, 8–12 reps at 80% 1RM	24	CLAMP	DXA	5 (Fair)
Guelfi et al. (2013)	Inactive overweight and obese male	49 ± 7	3×/week, three sets of 10 reps at 75%–85% of estimated 1RM	12	OGTT	DXA	5 (Fair)
Hurlbut (2002)	Healthy, younger, and older individuals	Male: young 25 ± 3.2; older: 71 ± 6.9 Female: young: 26 ± 3; older: 68 ± 6	3×/week, 5–15 reps dropsets	24	OGTT	DXA	5 (fair)
Hurley et al. (1988)	Untrained middle-aged male	44 ± 3	3–4×/week, 15–20 reps/8–12 reps	16	OGTT	Hydrodensitometry	5 (Fair)
Ismail et al. (2019)	Overweight male	36 ± 8	3×/week from 80% 1RM to volitional failure	6	OGTT	Air-displacement plethysomography	4 (Fair)
Ivey and Ryan (2014)	Stroke survivors with hemiparetic gait	65 ± 6.3	3×/week, two sets 10–15 reps close to muscle failure	12	OGTT	DXA	4 (fair)
Joseph et al. (1999)	Healthy older individuals	60 ± 3	F: 2×/week, two sets for eight reps at 80% 1RM; third set at failure or 12 rep.	12 weeks	OGTT	Hydrodensitometry	7 (Good)
Karelis et al. (2016)	Individuals who underwent renal transplantation	Control: 39.4 ± 8 RT: 45.3 ± 14	4×/week, three sets of 10 reps at 80% of 1RM	16	OGTT	DXA	7 (Good)
Kitamura et al. (2003)	Older, healthy individuals	69 ± 2.6	3×/week, two sets of 20 reps 30%–60% of 1RM (to ensure muscle failure)	12	CLAMP	DXA	7 (Good)

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Table 2. (*concluded*).

Study	Main characteristics of participants	Age (years)	RT overview	Duration (weeks)	Glucose metabolism	BC measurement	Quality rating/11 (PEDro scale)
Layne et al. (2011)	Participants with metabolic syndrome and healthy participants	Healthy: 36.4 ± 12.2 Metabolic syndrome: 45.0 ± 8.6	5×/week, three sets of 10 reps on phase I and five sets of three reps on phase II	8	CLAMP	Air-displacement plethysomography	5 (Fair)
Lott et al. (2001)	Healthy elderly participants	Male: 70 ± 3 Female: 69 ± 3	3×/week, one–two sets of 15 reps at muscle failure	24	OGTT	DXA	3 (Poor)
Malin et al. (2013)	Young female with different body compositions	High fat mass: 21.0 ± 2.8 Normal fat mass: 21.9 ± 2.3 Control: 20.9 ± 1.6	3x/week, three sets at 60% of 1RM for 8–12 reps	7	OGTT	DXA	6 (Good)
Miller et al. (1994)	Healthy male college students	Not mentioned	3×/week, circuit format	16	OGTT	Hydrodensitometry	4 (Fair)
Miller et al. (1984)	Healthy older male	58 ± 3.3	3×/week, three sets of eight reps (intensity not defined)	10	OGTT	Hydrodensitometry	3 (Poor)
Nybo et al. (2010)	Healthy, untrained male	Control: 30 ± 6.6 RT: 36 ± 5.7	2×/week, three–four sets for 12–16 RM	12	OGTT	DXA	5 (Fair)
Poehlman et al. (2000)	Premenopausal female	Control: 28 ± 4 RT: 28 ± 3	3×/week, three sets of 10 reps at 80% of 1RM	28	CLAMP	DXA	5 (Fair)
Reynolds et al. (2004)	Older individuals with hypertension	67 ± 2	3×/week, two sets of 10–12 reps	16	CLAMP	DXA	4 (Fair)
Roberts et al. (2013)	Sedentary, overweight, or obese young male	Control: 21.9 ± 1.8 RT: 21.5 ± 2.3	3×/week, two sets at 6–12 reps to muscle failure	12	OGTT	DXA	6 (Good)
Rosa et al. (2021)	Hemodialysis patients	Control: 52 ± 17 RT: 53 ± 13	3×/week, three sets at 8–12 reps, 5–6 on RPE (Borg scale)	24	OGTT	DXA	6 (Good)
Robinson et al. (2017)	Young and older healthy adults	Men: 43 ± 2.6 Women: 46 ± 2.3	4×/week, two–four sets at 8–12 reps for 8–12 exercises	12	CLAMP	DXA	6 (Good)
Russell et al. (2017)	Adults with T2D	52 ± 8.3	3×/week, one set for 6–15 reps at muscle failure	6	OGTT	DXA	4 (Fair)
Ryan et al. (2001)	Insulin resistant older individuals	69 ± 4	3×/week, one–two sets of 15 (dropsets)	24	CLAMP	DXA	5 (Fair)
Ryan et al. (1996)	Healthy post-menopausal female	58 ± 5	3×/week, one–two sets of 15 (dropsets)	24	CLAMP	DXA	5 (Fair)
Smutok et al. (1994)	Men with abnormal glucose regulation	Control: 57 ± 9 RT: 52 ± 11	3×/week, two sets of 12–15 at RM	20	OGTT	Hydrodensitometry	6 (Good)
Smutok et al. (1993)	Middle-aged male at high risk for coronary heart disease	Control: Age: 50 ± 8 RT: 48 ± 12	3×/week, two sets of 12–15 at RM	20	OGTT	Hydrodensitometry	5 (Fair)
Zachwieja et al. (1996)	Healthy, sedentary older male	66.3 ± 1.8	3×/week, Four sets of 4 to 10 at 75% to 90% RM	16	IVGTT (3 h)	Hydrodensitometry	9 (Excellent)

Note: Data are presented as mean ± SD (when applicable). *n*: number of participants; BC: body composition; BMI: body mass index (kg/m^2); CLAMP: hyperinsulinaemic-euglycemic clamp; DXA: dual X-ray absorptiometry; MRI: magnetic resonance imaging; FITT-VP: Frequency, Intensity, Type, Time—Volume (defined as total number of sets performed per week), Progression; GM: glucose metabolism; IVGTT: intravenous glucose tolerance test; OGTT: oral glucose tolerance test; RM: repetition maximum; T2D: type 2 diabetes; RCT: randomized controlled trial; RT: resistance training; Reps: repetitions; sets: series; RPE: rate of perceived exertion.

Table 3. Assessment of certainty of evidence for each outcome assessed.

Outcome	Relative effect [95% CI]	Number (N) of participants (studies)	GRADE	Comments
Skeletal muscle hypertrophy/increased FFM; pre–post intervention	0.24 [0.10; 0.39]	1054 (7 RCTs)	⊕⊕⊕	Evidence coming mostly from single-arm or parallel-arm studies
Skeletal muscle hypertrophy/increased FFM; analyses on studies including a control group	0.42 [−0.04; 0.88]	157 (7 RCTs)	⊕⊕	Evidence coming mostly from RCTs Small sample size (imprecision) Overall moderate risk of bias Inconsistency
Change in 2 h glucose; analyses on pre–post intervention	−0.30 [−0.50; −0.11]	439 (6 RCTs)	⊕	Evidence coming mostly from single-arm or parallel-arm studies Overall moderate risk of bias Imprecision
Change in 2 h glucose; analyses on studies including a control group	0.40 [−1.01; 0.22] (non-significant)	145 (6 RCTs)	⊕⊕	Evidence coming mostly from RCTs Small sample size Overall moderate risk of bias Inconsistency
Change in glucose AUC; analyses on pre–post intervention	−0.40 [−0.66; −0.14]	452 (5 RCTs)	⊕	Evidence coming mostly from single-arm or parallel-arm studies Overall moderate risk of bias Imprecision
Change in glucose AUC; analyses on studies including a control group	−1.3 [−3.50; 0.99] (non-significant)	153 (5 RCTs)	⊕⊕	Evidence coming mostly from RCTs Small sample size Overall moderate risk of bias Inconsistency
Changes in insulin sensitivity; analyses on pre–post intervention	0.38 [0.20; 0.56]	222 (1 RCT)	⊕	Evidence coming mostly from single-arm or parallel-arm studies Overall moderate risk of bias Imprecision

Note: CI: confidence interval; number (N) of participants: total number of participants from studies on a given outcome; (studies): number of studies that qualifies as randomized controlled trials; FFM: fat-free mass; RCT: randomized controlled trial.
GRADE working Group grades of evidence rating for certainty of evidence: ⊕: Very low; ⊕⊕: Low; ⊕⊕⊕: Moderate; ⊕⊕⊕⊕: High.
High: We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low: Our confidence in the effect estimate is limited; The true effect may be substantially different from the estimate of the effect.
Very low: We have very little confidence in the effect estimate; The true effect is likely to be substantially different from the estimate of effect.

Fig. 2. Funnel plot, using data from studies with oral glucose tolerance test 2 h glucose outcome.

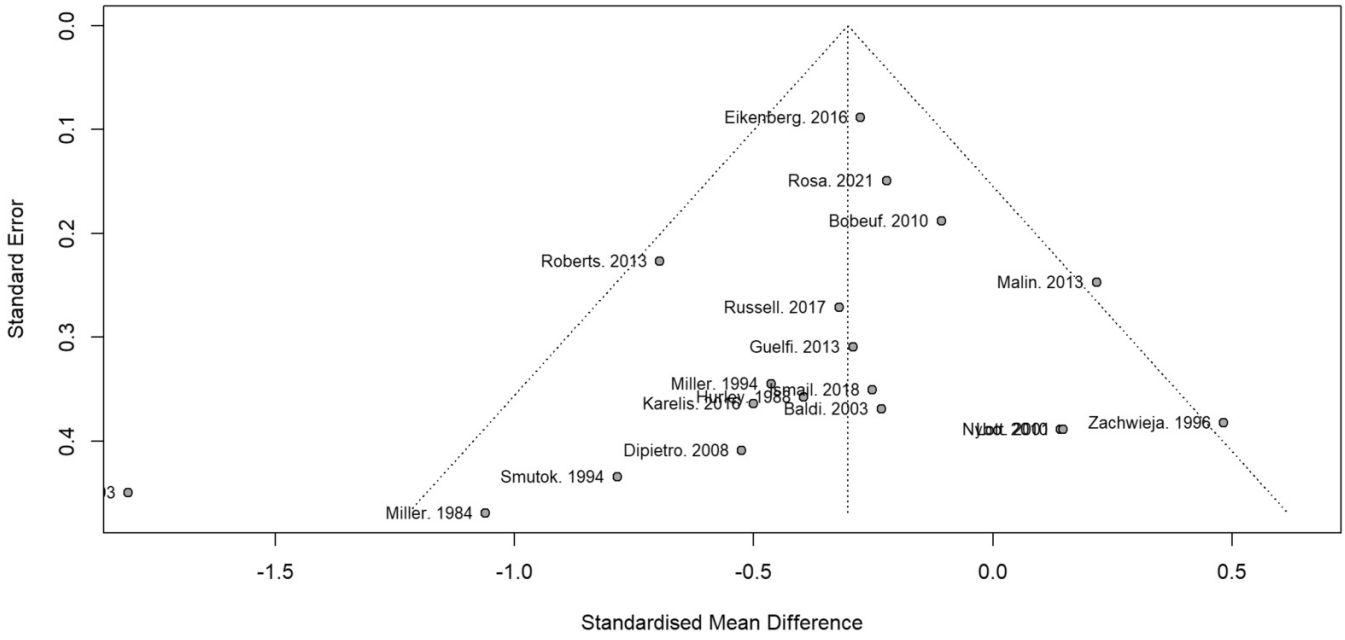
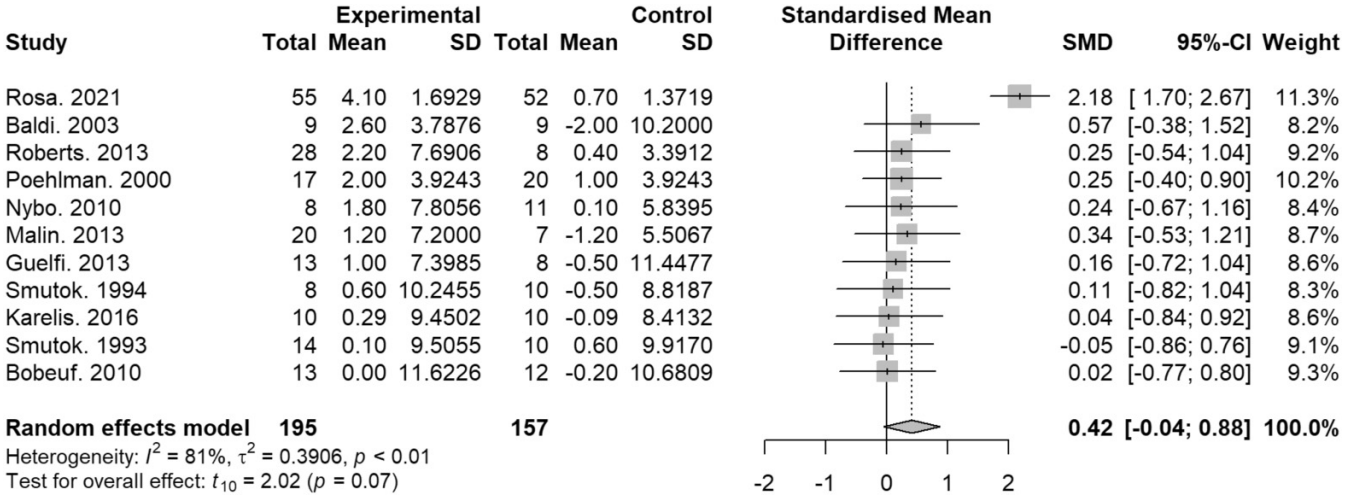


Fig. 3. Standardized mean differences (SMDs) in fat-free mass changes with resistance training interventions (compared with control group). Results are shown as SMD [95 confidence interval (CI)%]; I^2 for heterogeneity; Total: number of participants.



sensitivity derived from CLAMP studies, also with a moderate effect size (0.38 [0.20; 0.56]; $p < 0.01$; $I^2 = 18.1\%$).

4.4.2. Analyses on studies including a control group

Figure 4 displays results on analyses for 2 h glucose in studies with a control group. When compared to control groups, there was no significant difference in 2 h glucose, nor in glucose AUC from baseline in RT intervention groups (−0.40 [−1.01; 0.22], $p = 0.17$; $I^2 = 86\%$; −1.3 [−3.50; 0.99], $p = 0.21$, $I^2 = 96.2$). No analysis on intervention versus control group could be performed for studies on CLAMP, as only one study comprised a control group.

4.5. Relationship between fat-free mass quantitative changes, glucose homeostasis and confounding factors

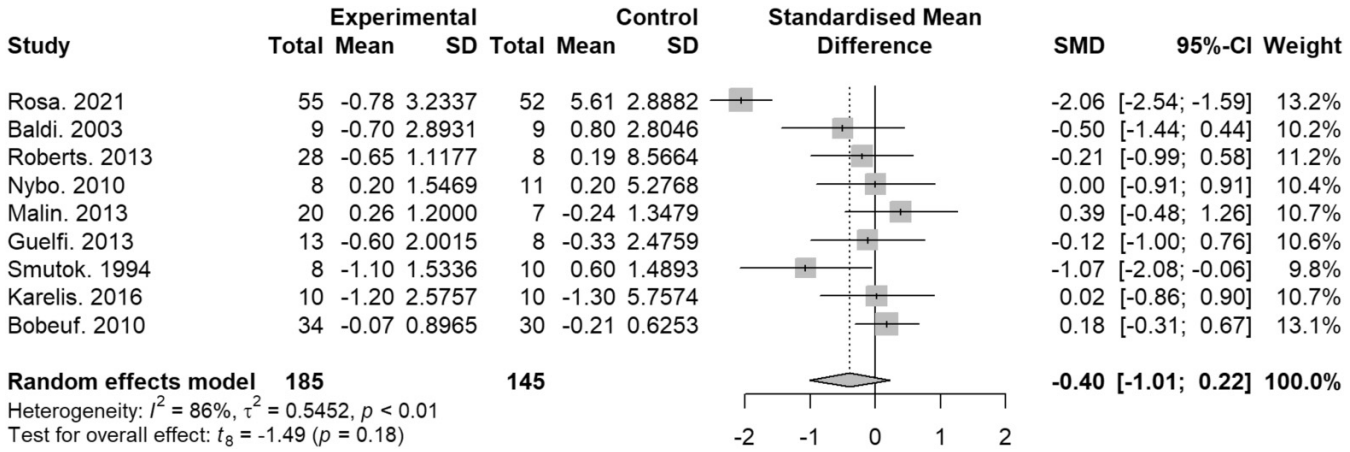
A direct association between quantitative changes in FFM and OGTT-derived glucose tolerance could not be inferred from visual examination of caterpillar plots (Figs. 3 and 4). Indeed, if FFM increase or decrease was the major contributor to improved glucose tolerance or insulin sensitivity, a linear pattern would have emerged, which was not the case (see Fig. S1 for an example with fictional data). Meta-regressions were performed on all intervention (RT) groups to determine whether pooled effect size estimate for glucose homeostasis outcomes were influenced by absolute FFM changes and other confounding variables. FFM increases were found to be a positive moderator for glucose AUC improvement ($\beta = -0.28$; [0.0981; −2.934]; $p = 0.011$; amount of heterogeneity accounted for in model: 51.7%, $I^2 = 61.2\%$), but not for 2 h glucose (OGTT) or insulin sensitivity (CLAMP). Because Rosa et al. (2021) was as a statistical outlier (≥3 SDs from mean), we conducted the same analysis without those results as a sensitivity analysis. Model results then showed no significant effect of FFM increases as a moderator of glucose AUC improvement ($\beta = -0.0013$; [0.15; 0.009]; $p = 0.99$, amount of heterogeneity accounted for in model: 0.0%, $I^2 = 44.5\%$). We conducted additional mixed-effects model analyses including all studies to examine the contribution of proportion of females in study sample, training intervention duration, FM loss, total volume performed, changes in VO_2max/VO_2peak, and sample's average age and we found no significant effect neither for glucose homeostasis nor FFM increase. Finally, subgroup analyses revealed no significant differences when separating studies with and without T2D participants (p value for subgroup differences ranging from 0.17 to 0.82). Results from meta-regression using age as a moderator and subgroup analysis separating studies with T2D participants can be found in supplementary material.

5. Discussion

This meta-analysis examined FFM and glucose metabolism adaptations in response to prolonged RT in adults, with two aims: first, assess the magnitude of FFM increase and glucose homeostasis improvement in response to RT interventions, and second, to examine the relationship between RT-induced FFM gain and improvements in glucose homeostasis. Albeit the role of RT on whole-body glucose metabolism being well-established, very few studies have addressed with a systematic approach the complex issue of the impact of RT-induced increase in FFM on glucose homeostasis. These results are of important significance with regard to the actual effectiveness of RT for increasing FFM (which can be inferred as Skm hypertrophy for reasons previously stated) and improving glucose metabolism.

Analyses on single and parallel-arm studies revealed a significant effect of RT for increasing FFM, which is in accordance with results of several previous investigations and meta-analyses (Schoenfeld et al. 2017*b*; Damas et al. 2018) and with the body of literature that has demonstrated the role of RT in muscle protein synthesis and consequent muscle hypertrophy (Witard et al. 2021). However, analyses on studies com-

Fig. 4. Two hour glucose standardized mean differences versus control group (oral glucose tolerance test). Results are shown as standardized mean difference (SMD) [95 confidence interval (CI)%]; I^2 for heterogeneity. Total: number of participants.



prising a control group did not reach statistical significance. A note of caution is due here since the assessment of certainty of evidence (GRADE) revealed low certainty when analyses were restricted to controlled studies only, mostly due to high heterogeneity and imprecision, whereas within-studies analyses were assessed as having a high level of certainty of evidence. It is also possible that such inconsistency between mixed-effects models is due to our inclusion/exclusion criteria that were not selected to analyze the sole effect of RT on FFM and therefore, may have limited our sample size. Results from analyses on single and parallel-arm studies only showed a significant impact of RT on glucose tolerance measured with the OGTT, which was also not the case in analyses including controlled studies. These results came as surprising, since improved glucose tolerance following RT has been repeatedly documented since the 1980s (Yki-Jarvinen and Koivisto 1983; Miller et al. 1984, 1994; Eriksson et al. 1997). However, FFM increase was only uncovered as a significant contributor of glucose AUC improvement in response to RT, with sensitivity analysis resulting in no significance effect once the main statistical outlier was removed. Indeed, some studies included in the current meta-analysis reported minor to no change in glucose tolerance after RT intervention albeit significant Skm/FFM increase (Zachwieja et al. 1996; Hurlbut 2002; Guelfi et al. 2013; Malin et al. 2013). This is paradoxical to other studies reporting superior glucose tolerance among individual with higher Skm mass cross-sectionally (Kalyani et al. 2012), while also corroborating others reporting no relationship between glucose tolerance and Skm mass (Kuk et al. 2008). Although the OGTT and the CLAMP techniques both measure glucose metabolism under metabolic challenges conditions, one must consider the major differences in their basic assumptions and methodologies when comparing their respective outcomes. While the CLAMP has been accepted as the most rigorous method for determining insulin sensitivity (Otten et al. 2014) the OGTT is rather a measurement of whole-body glucose tolerance, often used as a tool for T2D and gestational diabetes diagnosis (Jagannathan et al. 2020). Hence, results from the OGTT cannot be used as a direct surrogate of peripheral glucose uptake (Antuna-Puente et al. 2011). Importantly, certainty of evidence from analyses on studies with CLAMP must be considered very low, as only one study included a control group.

Results from meta-regression revealed no evidence of a direct influence of FFM quantitative changes on insulin sensitivity, albeit the significant effect of RT on CLAMP-derived insulin sensitivity, as previously suggested (Dela and Kjaer 2006). In the current meta-analysis, we have also attempted to examine whether training parameters, such as total volume and the duration of the intervention, might explain FFM and glucose metabolism adaptations in response to prolonged RT. Unexpectedly, none of these variables was found as a significant contributor of FFM increases, nor of glucose homeostasis improvement. These results came as surprising, especially given the documented role of training volume in Skm adaptations in response to RT (Brandenburg and Docherty 2006). Furthermore, meta-regression analyses did not support biological sex as a contributor of FFM increases, which indicates no significant sex differences in FFM quantitative changes in response to RT within included studies. These results support those from a recent meta-analysis of Roberts et al., which highlighted that FFM and strength are likely to adapt to RT comparably in both sexes, regardless of sex differences at baseline (Roberts et al. 2020). Nonetheless, FM changes were not found as contributors of improved glucose homeostasis in response to RT.

Age-related muscle loss, or sarcopenia, has long been suggested as one of many potential culprits of T2D development in longitudinal studies (Haran et al. 2012; Cleasby et al. 2016). Alternatively, qualitative changes in Skm tissue, such as decreased mitochondrial content (Bellanti et al. 2021), have also been pointed out as a potential contributor to both sarcopenia and decreased Skm glucose uptake. Moreover, older individuals are characterized by a blunted hypertrophic response to RT (Peterson et al. 2011). Therefore, to explore potential bias due to our large open inclusion criteria regarding age, we performed meta-regression analyses with age as a cofounding factor and found no significant effect of age as a moderator of

glucose homeostasis nor FFM changes. Finally, we performed sub-groups analyses (mixed-effects models with studies including T2D participants as a subgroup) and found no significant subgroup differences. Therefore, a different response to RT-induced increase in FFM and glucose homeostasis in individuals with a T2D diagnosis cannot be concluded within the current systematic review and meta-analysis.

5.1. Strengths and limitations

This is the first meta-analysis highlighting the effect of RT on whole-body glucose homeostasis adaptations and the role of FFM changes in this context. Our broad inclusion criteria for population sample may help generalize our results to various adult populations, as we have included studies in participants with various age and health status (young and older healthy individuals, individuals with metabolic syndrome, T2D individuals, etc.). Furthermore, only including studies using metabolic challenges provides stronger evidence from a mechanistic but also clinical point of view. However, the majority of studies included do not include a control (no exercise) group, and hence do not meet criteria for RCTs. Though, as previously mentioned, conducting robust exercise trials comprises major challenges, such as study environment, sample size, and ethical considerations (Hecksteden et al. 2018) and may be considered as a limit to most exercise intervention studies. At the study level, factors that may generate limitations in the interpretation and application of results merit attention. Notably, an important number of selected studies do not provide essential details with regards to exercise prescription and progressive overload (Bushman 2018). Likewise, a limited proportion of studies have used the consensus on exercise reporting template (CERT) guidelines (Slade et al. 2016) when reporting their RT protocol, and only a few provided details pertaining to progressive overload. This lack of robustness in reporting RT prescription might explain, in part, our inability to identify significant moderators of FFM and glucose homeostasis changes among RT prescription parameters. Finally, although DXA-scan has long been accepted as the "reference standard" method in clinical settings for body composition assessment (Buckinx et al. 2018), and was univocally the most utilized body composition assessment tool in the current meta-analysis, it does not, in fact, directly measures Skm mass (Evans et al. 2019). Rather, DXA-scan provides fat and bone tissue mass estimation according to their level of attenuation of X-rays, and FFM therefore represents the remainder, i.e., the sum of lipid-free tissue—based on the assumptions that X-ray beam attenuation is proportional to a tissue mass and that fat, bone, and lean tissue have a fixed density (Proctor et al. 1999; Toombs et al. 2012; Messina et al. 2020). Recently, a study of Tavoian et al. investigated the concordance in measures of percentage changes of Skm/FFM between DXA-scan and MRI in RT interventions studies and found a positive, but modest association between MRI and DXA-scan, with DXA-scan showing higher heterogeneity than MRI, as well as overestimating changes in FFM in a majority of cases (Tavoian et al. 2019). If the debate on the effect of Skm hypertrophy on glucose homeostasis is to be moved forward, a better understanding of the relationship between quantitative changes in Skm and glucose metabolism could be provided by clinical trials using more precise body composition assessment tools, such as MRI and CT-scan (Messina et al. 2020).

6. Conclusion

To the best of our knowledge, this is the first meta-analysis that examines the relationship between changes in FFM and glucose homeostasis improvements after prolonged RT in adult populations. One of the most significant findings is that while RT is a potent strategy to generate increases in FFM/Skm hypertrophy, glucose tolerance, and peripheral insulin sensitivity, these adaptations are likely independent from one another. Furthermore, these counterintuitive results were not explained by biological sex, age, nor training parameters. Our findings may inform practitioners using RT in clinical settings for T2D prevention or treatment of its concurrent, but perhaps independent effects on metabolic health and FFM quantity. In research settings, a greater focus on strictly controlled training parameters, such as volume, duration of intervention, and training protocol would lead to the development of evidence-based exercise recommendations aiming to improve glucose metabolism.

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Data availability

Data generated or analyzed during this study are available from the corresponding author upon request.

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Competing interests

The authors declare no conflicts of interest.

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Supplementary material

Supplementary data are available with the article at https://doi.org/10.1139/apnm-2024-0128.

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